Department of Artificial Intelligence and Machine Learning

Course Code: AI372IA **Course:** Stream Processing and Analytics

Project Synopsis

Project Title	Smart Agriculture Crop Health Monitoring
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1. Introduction and Relevance of the Project

Agricultural monitoring increasingly relies on weather observations, environmental data and satellite-derived indicators to understand conditions that may affect crop growth. This project applies real-time stream processing to agricultural risk monitoring, with an emphasis on reliable data handling and timely, interpretable alerts.

- **Background:** Environmental data can arrive continuously and may contain missing, delayed, duplicated or implausible observations. Analysing individual readings without temporal or weather context can produce misleading alerts.
- **Relevance:** Early identification of potentially adverse growing conditions can help farmers or agricultural analysts decide when a field warrants further inspection.
- **Current technologies:** Apache Kafka supports event-stream ingestion, while PySpark Structured Streaming enables scalable transformations and time-dependent analysis. Public weather APIs and, where feasible, satellite-derived vegetation indicators can provide environmental context.
- **Research gap addressed:** The project will investigate whether context-aware risk detection combined with explicit data-quality checks can reduce false alerts while preserving timely detection, and will measure pipeline resilience under controlled failures.
- **Scope:** The prototype will use public weather/environmental APIs and historical data for 10 virtual locations in Karnataka. Satellite-derived indicators are conditional on feasibility. No physical farm sensors or verified field-level crop-health labels are assumed.

2. Problem Statement and Objectives

Problem statement: To develop a fault-tolerant, real-time agricultural monitoring and decision-support system using Kafka and PySpark Structured Streaming to detect potential crop-stress risks through context-aware, data-quality-aware anomaly detection, generate evidence-based AI recommendations, and evaluate detection and system reliability against conventional baselines.

Objectives:

1. Develop a Kafka-centred pipeline for ingesting and processing agricultural environmental data.
2. Implement context-aware anomaly detection using temporal trends and available weather/environmental variables.
3. Identify missing, delayed, duplicated and implausible observations so data-quality problems can be distinguished from potential environmental risks.
4. Integrate an AI-assisted advisor that produces evidence-grounded explanations and recommendations while communicating uncertainty.
5. Build a dashboard showing risk alerts, supporting evidence, data freshness and stream health.
6. Compare the proposed detector with fixed-threshold and rolling-statistical baselines, and measure detection quality, latency, throughput and recovery under controlled tests.

3. Proposed Methodology / Approach

Data Source → Ingestion → Apache Kafka → PySpark Structured Streaming → Risk Detection → Storage and Dashboard

The system separates deterministic stream processing from the slower AI-assisted advisory stage. Incoming observations are processed by the streaming pipeline; the advisor is invoked for selected alerts rather than for every event.

- **Data collection:** A Python ingestion service will retrieve available weather/environmental observations from public APIs and load historical data for 10 virtual Karnataka locations. Satellite-derived vegetation indicators may be added if access and processing are feasible.
- **Preprocessing and Kafka ingestion:** The ingestion service will normalize fields, validate schemas, attach timestamps, and publish records to appropriate Kafka topics. Missing or malformed values will be flagged rather than silently treated as valid observations.
- **Stream processing:** PySpark Structured Streaming will compute temporal features and contextual summaries, handle event-time and late-arriving data as appropriate, and maintain recent observations for risk assessment.
- **Data-quality assessment:** Checks will identify missing, duplicated, delayed or implausible records. The detector will use these quality indicators to avoid interpreting every unusual reading as a genuine crop-stress event.
- **Context-aware risk detection:** The proposed detector will combine temporal trends with available weather and environmental context to score potential risk. Fixed-threshold and rolling-statistical methods will serve as comparison baselines.
- **AI-assisted advisory:** For selected alerts, an AI component will use the alert evidence and agricultural guidance to explain possible risks and suggest verification steps. It will state uncertainty and will not claim a definitive disease diagnosis from weather or modelled soil moisture alone.
- **Storage and visualisation:** PostgreSQL will store processed observations, risk alerts and evaluation records. A dashboard will present field-level alerts, time-series context, evidence, data freshness and streaming-system health.
- **Fault tolerance:** The implementation will use appropriate Kafka and Spark recovery mechanisms, such as checkpointing and restart handling, and will test selected failures, delayed or malformed data, and recovery behaviour.

Implementation steps: (1) Data review and exploratory analysis, (2) API/historical-data ingestion, (3) Kafka topics and streaming pipeline,
(4) data-quality checks and detection baselines, (5) proposed detector and AI advisory, (6) dashboard and storage, and (7) comparative evaluation and fault-injection tests.

4. Expected Outcomes

- **Prototype:** A locally deployable agricultural-risk monitoring pipeline connecting data ingestion, Kafka, PySpark Structured Streaming, PostgreSQL, an AI-assisted advisor and a dashboard.
- **Risk alerts:** Field/location-level alerts that display supporting environmental evidence, data-quality status and uncertainty. Alerts will indicate potential risk, not verified crop disease.
- **Comparative evaluation:** Measured comparison of the proposed detector against fixed-threshold and rolling-statistical baselines using precision, recall, false-alert rate and detection delay where suitable labelled test scenarios are available.
- **Streaming performance:** An evaluation of throughput, end-to-end processing latency, and resource use under selected data loads.
- **Reliability assessment:** Documented results for selected failure-injection scenarios, including recovery time and any missed or duplicated events observed.
- **Research deliverables:** A reproducible evaluation report documenting the dataset limitations, test scenarios, measured results and limitations. Real-world agronomic benefits will not be claimed without validation against ground-truth field observations.